



Egypt-Japan University of Science and Technology
Computer Science and Engineering Department

CSE 429: Computer Vision and Pattern Recognition

Project Proposal

Arabic Handwritten Text Recognition using TrOCR

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Problem Statement

Arabic handwritten text recognition is still a challenging problem because Arabic writing is cursive, characters change shape depending on their position in the word, and handwriting styles vary significantly between writers. Traditional OCR systems often depend on CNNs, RNNs, and additional language models, which increases the complexity of the pipeline.

The goal of this project is to fine-tune the Transformer-based OCR model **TrOCR** for Arabic handwritten text recognition. The system input will be an image containing a single Arabic handwritten text line, and the output will be the corresponding digital Arabic text. The project aims to evaluate whether a large pre-trained Transformer OCR model can achieve strong performance on Arabic handwritten recognition tasks.

Approach

This project is based on the paper “*TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models*”. TrOCR is an end-to-end encoder-decoder Transformer model that combines a Vision Transformer encoder for image understanding with a Transformer decoder for text generation.

We will fine-tune the pre-trained handwritten TrOCR model using the **KHATT v1.0** dataset, which is a benchmark dataset for Arabic offline handwritten text recognition. Since the original TrOCR decoder is based on English language modeling, we modify the decoder to better support Arabic text generation by using the **bert-base-arabic** model as the decoder initialization. This modification is intended to improve the model’s understanding of Arabic vocabulary and linguistic patterns during text generation.

The implementation will be carried out using Python, PyTorch, and the Hugging Face Transformers library. The preprocessing pipeline will include image resizing, normalization, and tokenization of Arabic text labels. In addition, simple data augmentation techniques such as rotation and noise injection will be explored to improve model generalization.

Experiments and Expected Results

The experiments will evaluate the performance of the fine-tuned TrOCR model on the KHATT dataset and compare the results with the paper “*A Transformer-based Approach for Arabic Offline Handwritten Text Recognition*”, which also evaluates Transformer-based methods on Arabic handwritten text recognition.

The main evaluation metric will be **Character Error Rate (CER)**, since the task focuses on handwritten text-line recognition. Additional metrics such as **Word Error Rate (WER)** and **BLEU score** will also be reported.

The planned experiments include:

- Fine-tuning TrOCR on the KHATT dataset
- Evaluating the effect of data augmentation
- Comparing the obtained CER and WER with the baseline paper
- Evaluating the impact of replacing the original English decoder with an Arabic language decoder

A successful outcome for this project would be achieving a CER comparable to or lower than the reported baseline results. The experiments are expected to demonstrate the effectiveness of pre-trained Transformer-based OCR models for Arabic handwritten text recognition and the importance of using an Arabic-aware decoder for text generation.